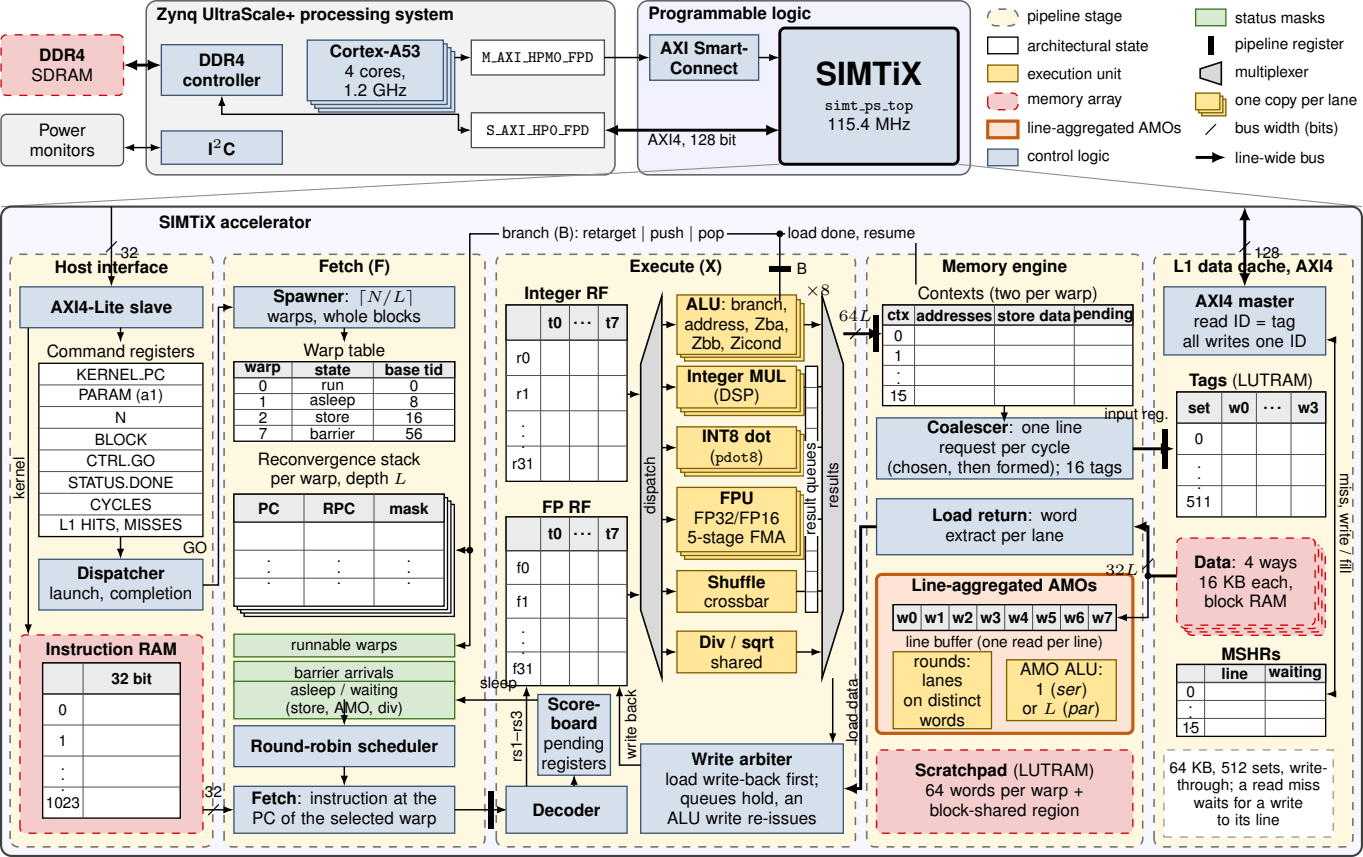

Zynq UltraScale+ processing system
DDR4 SDRAM
DDR4 controller
Cortex-A53 4 cores, 1.2 GHz
M_AXI_HPM0_FPD
S_AXI_HP0_FPD
Power monitors
I2C
Programmable logic
AXI Smart-Connect
SIMTiX
simt_ps_top
115.4 MHz
AXI4, 128 bit
pipeline stage
architectural state
execution unit
memory array
line-aggregated AMOs
control logic
status masks
pipeline register
multiplexer
one copy per lane
bus width (bits)
line-wide bus
SIMTiX accelerator
branch (B): retarget | push | pop
load done, resume
32
128
Host interface
AXI4-Lite slave
Command registers
KERNEL_PC
PARAM (a1)
N
BLOCK
CTRL.GO
STATUS.DONE
CYCLES
L1 HITS, MISSES
kernel
GO
Dispatcher launch, completion
Instruction RAM
32 bit
0
1
1023
Fetch (F)
Spawner: ⌈N/L⌉ warps, whole blocks
Warp table
warp
state
base tid
0
run
0
1
asleep
8
2
store
16
7
barrier
56
Reconvergence stack per warp, depth L
PC
RPC
mask
runnable warps
barrier arrivals
asleep / waiting (store, AMO, div)
Round-robin scheduler
Fetch: instruction at the PC of the selected warp
32
Execute (X)
B
Integer RF
t0
t7
r0
r31
FP RF
f0
f31
ALU: branch, address, Zba, Zbb, Zicond
Integer MUL (DSP)
INT8 dot (pdot8)
FPU FP32/FP16 5-stage FMA
Shuffle crossbar
Div / sqrt shared
×8
64L
dispatch
result queues
results
load data
sleep
Score-board pending registers
write back
rs1–rs3
Decoder
Write arbiter
load write-back first; queues hold, an ALU write re-issues
Memory engine
Contexts (two per warp)
ctx
addresses
store data
pending
Coalescer: one line request per cycle (chosen, then formed); 16 tags
input reg.
Load return: word extract per lane
32L
Line-aggregated AMOs
w0
w1
w2
w3
w4
w5
w6
w7
line buffer (one read per line)
rounds: lanes on distinct words
AMO ALU: 1 (ser) or L (par)
Scratchpad (LUTRAM)
64 words per warp + block-shared region
L1 data cache, AXI4
AXI4 master
read ID = tag
all writes one ID
Tags (LUTRAM)
set
w0
w3
0
511
miss, write / fill
Data: 4 ways 16 KB each, block RAM
MSHRs
line
waiting
15
64 KB, 512 sets, write-through; a read miss waits for a write to its line